Probability & Expectation

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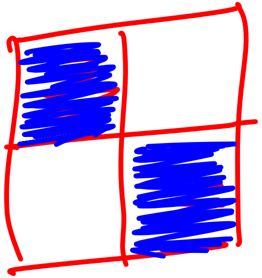
Goal

- Probability basics (+ examples)
- Conditional probability (+ examples)
- Dependent vs independent events (+ examples)
- Expectation (+ examples)
  - Linearity of expectation
  - Problems
- Problems on bounded probability

What is Probability?

Finding out how likely an event will occur.



$2/4$

$$\text{Probability of an event} = \frac{\text{Total number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$0 \leq P(\text{event}) \leq 1$

Examples

- Probability of getting a head on tossing a fair coin. → $1/2$
- Probability of getting at least 1 head on tossing 3 coins.
- Probability of picking a green ball from a bag 3 red balls, 2 blue balls, 5 green balls. → $5/10 = 1/2$
- Probability of getting a sum of 7 on tossing 2 dice. → $1/6$

✓ HHH
✓ HHT
✓ HTH
✓ HTT
✓ THH
✓ THT
✓ TTH
TTT

$\longrightarrow$ fav $\rightarrow$ 7
total = 8

$\rightarrow \underline{\underline{7/8}}$

P(event $\underline{\underline{\text{happening}}}$) = 1 − P(event not happening)

Total outcomes = fav. outcomes + unfav. outcomes

$$\frac{fav}{total}$$

$$fav = total - unfav$$

$$\frac{fav}{total} = 1 - \frac{unfav}{total}$$

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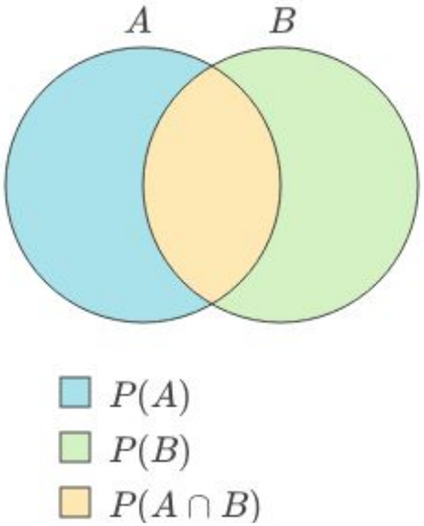
total = outcomes for 1st dice × outcomes for 2nd dice

$= 6 \times 6 = 36$

fav → {1, 6}, {2, 5}, {3, 4}, {4, 3},
{5, 2}, {6, 1} = 6

Conditional Probability

Conditional probability is the probability of an event A occurring given that another event B has already occurred. It is denoted as P(A|B) and calculated using the formula:



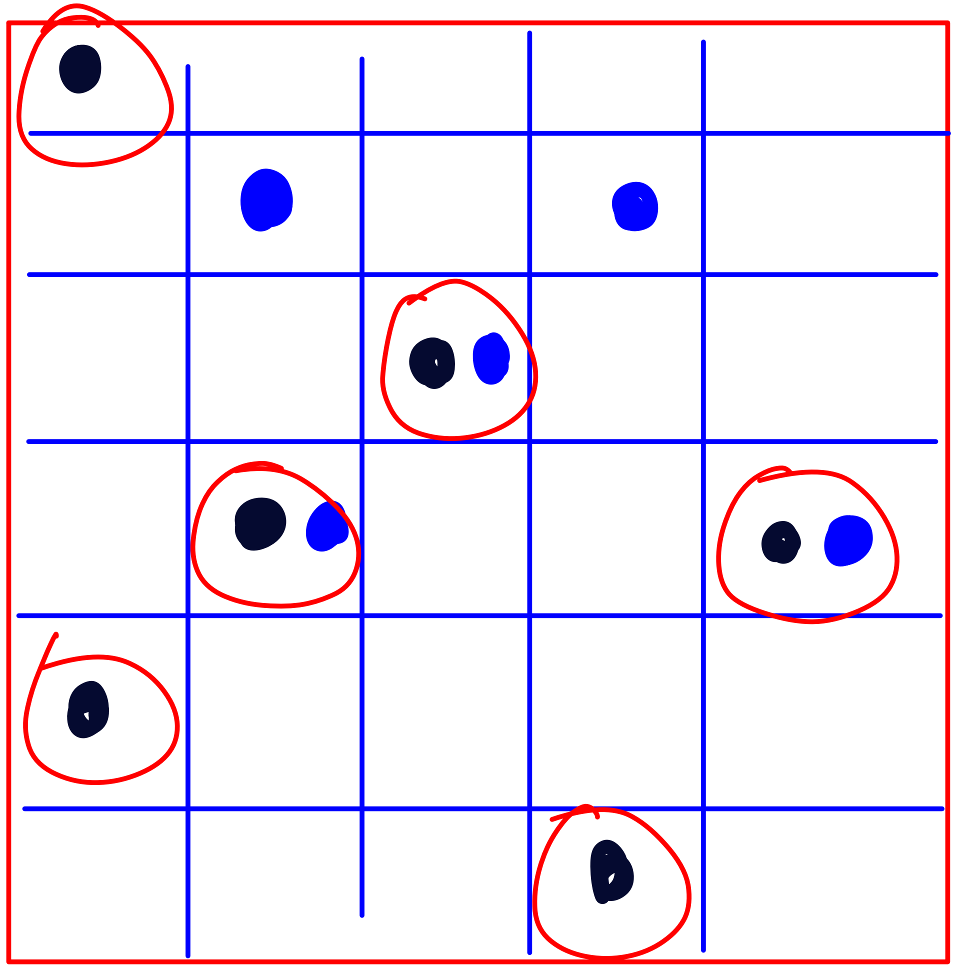
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Probability that A occurs given that B has already occured

$Event_1 \longrightarrow rain$ $\quad$ $Event_2 \rightarrow clouds$

$$P(event_1) \longrightarrow 2/7$$

$$P(\underline{event_1} \mid event_2)$$



cells = 30

$P(\bullet) \rightarrow 6/30$

$P(\bullet) = 5/30$

$P(\bullet \mid \bullet) = P(\bullet \cap \bullet) \,/\, P(\bullet) =$

$$P(\textcolor{blue}{\bullet} \mid \bullet) = 3/6 = 1/2$$

$$\underline{\underline{P(\textcolor{blue}{\bullet} \cap \bullet)}} = 3/30$$

$$\underline{\underline{P(\bullet)}} = 6/30$$

$$P(\textcolor{blue}{\bullet} \mid \bullet) = \frac{P(\textcolor{blue}{\bullet} \cap \bullet)}{P(\bullet)}$$

Example

Given a deck of of 52 cards. What is the probability of drawing a King, given that the card drawn is a face card?

4/12

- Total face cards: {Jack, Queen, King} for 4 suits = 12 cards.
- Favorable outcomes: 4 Kings.
- Conditional probability:

4/52 12/52

a. P(King | Face card) = P(King and Face card) / P(Face card)

t1 — 13 → A, K, Q, J, 2, 3, 4, 5, 6, 7, 8, 9, 10

t2 — 13

t3 — 13

t4 — 13

Event$_1$		Event$_2$
Rain	independent	TV
100% in Math exam in class 10	dependent	You got a 99%ile in IIT JEE

$$P(A|B) \neq P(A)$$

$$P(A|B) = P(A) \quad (\text{independent})$$

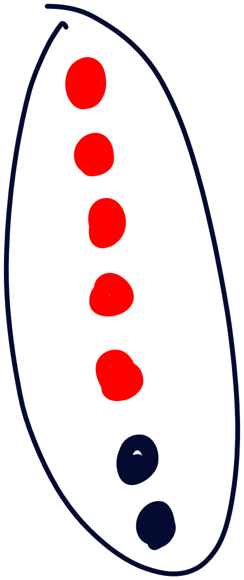
$$P(A \cap B) = P(A) \cdot P(B) \quad (\text{independent})$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

$$\hookrightarrow P(A \cap B) = P(A) \cdot P(B)$$

Dependent vs Independent Events

Independent Events	Dependent Events
Events where the outcome of one does not affect the other.	Events where the outcome of one affects the probability of the other.
$P(A \cap B) = P(A) \cdot P(B)$	$P(A \cap B) = P(A) * P(B \mid A)$
No conditional relationship exists.	Events are conditionally related.
Tossing two coins.	Drawing 2 balls without replacement



$$P(\text{1st ball} = \text{black}) = 2/7$$

$$P(\text{1st ball} = \text{red}) = 5/7$$

$$P(\text{2nd ball is red} \mid \text{1st ball was red}) = \underline{4/6}$$

$$P(\text{2nd ball is red} \mid \text{1st ball is black}) = 5/6$$

What is Expectation

If every outcome of an event had a value and a probability to occur, what is the average value we can expect from that event. Expectation is simply a weighted average.

f (outcome)

$$E[X] = \sum_i x_i f(x_i)$$

outcome

$$\text{Coin} \to H \to 10\ \text{Rs}$$
$$\text{Coin} \to T \to 20\ \text{Rs}$$

$$= \frac{10+20}{2} = 15$$

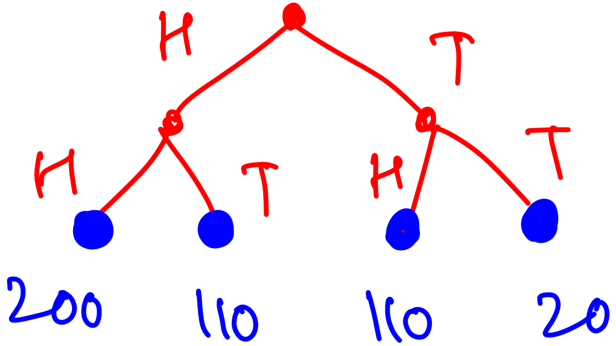
$$\text{Coin} \xrightarrow{3/4} H \to 10\ \text{Rs}$$
$$\text{Coin} \xrightarrow{1/4} T \to 20\ \text{Rs}$$

$$\frac{10+10+10+20}{4} = \frac{50}{4}$$

$$= 12.5$$

Examples

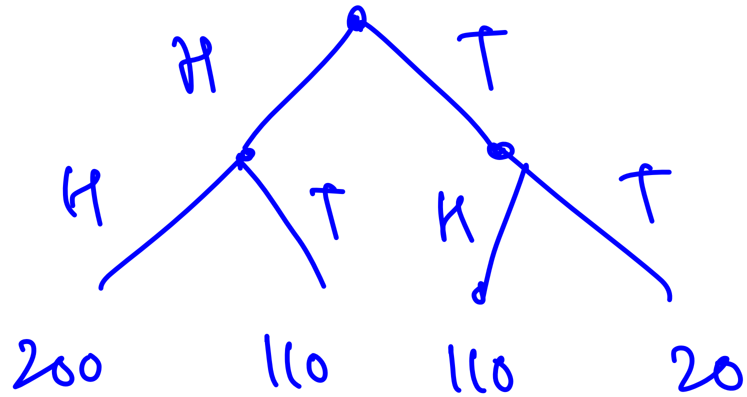
- Expected money that can be won on tossing a coin. Getting a head => Rs. 100. Getting a tail => Rs. 10.
- Expected sum of numbers on throwing ~~2 dice~~. the coin twice



$$\frac{440}{4} = 110$$

$$= 200 \cdot 1/4 + 110 \cdot 1/4 + 110 \cdot 1/4 + 20 \cdot 1/4$$

$$\left.\begin{array}{l} H \rightarrow 100 \\ T \rightarrow 10 \end{array}\right\} \quad 2 \text{ coin tosses}$$

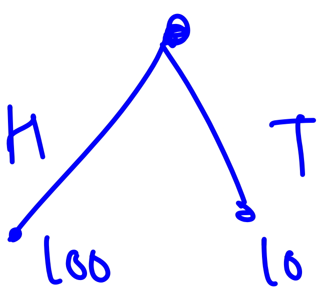


$$200 \cdot \frac{1}{4} + 110 \cdot \frac{1}{4} + 110 \cdot \frac{1}{4} + 20 \times \frac{1}{4}$$

$$= 110$$

E(total money on 1st + 2nd throw)

$$= \underbrace{E(\text{1st throw})}_{55} + \underbrace{E(\text{2nd throw})}_{55}$$



$$= 100 \cdot \frac{1}{2} + 10 \cdot \frac{1}{2} = \underline{\underline{55}}$$

Linearity of Expectation

$$\overset{x}{} \quad \overset{y}{}$$

$$E[X + Y] = E[X] + E[Y]$$

$$E[aX + b] = aE[X] + b$$

E(2 · coin toss) = 2 · E(coin toss)

$$\text{Coin} \to H \to 1 \qquad \text{Coin} \to T \to 2$$

$$\text{Dice} \to 1,\ 2,\ 3,\ 4,\ 5,\ 6$$

$E(\text{coin toss} + \text{dice throw})$

2	3	4	5	6	7
H, 1	H, 2	H, 3	H, 4	H, 5	H, 6
T, 1	T, 2	T, 3	T, 4	T, 5	T, 6
3	4	5	6	7	8

$$= \sum \text{outcome} \cdot p(\text{outcome})$$

$E(\text{coin toss} + \text{dice throw})$

$= E(\text{coin toss}) + E(\text{dice throw})$

$1.5 + 3.5$

$= \underline{\underline{5}}$

① H   ② T

$1/2$ $1/2$

$1, 2, 3, 4, 5, 6$

$$\frac{1+2+3+4+5+6}{6}$$

$$= \frac{6 \cdot 7}{2} \cdot \frac{1}{6} = 3.5$$

Problem 1:

What is expected number of throws after which you will get a head

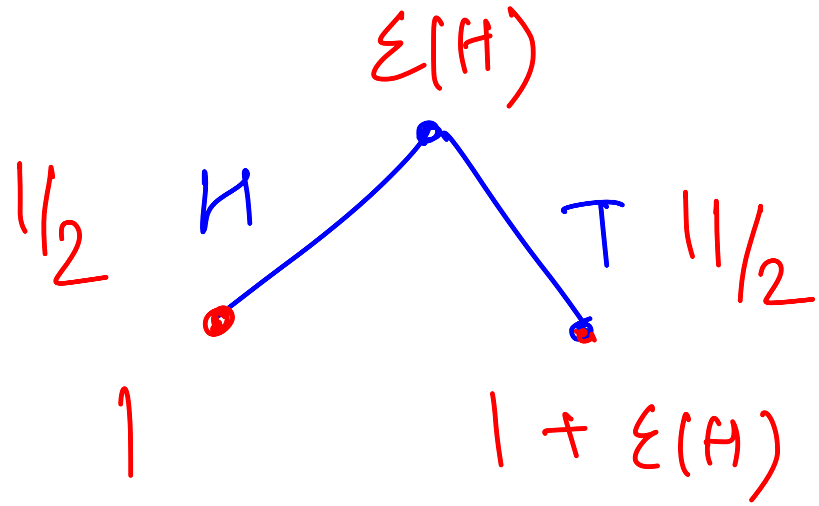
E(coin tosses to get a H)

$= \sum$ outcome $\cdot\ p$ (outcome)

$= 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} - \cdots\cdots$

$= \sum_{i=1}^{\infty} i \cdot 2^{-i}$

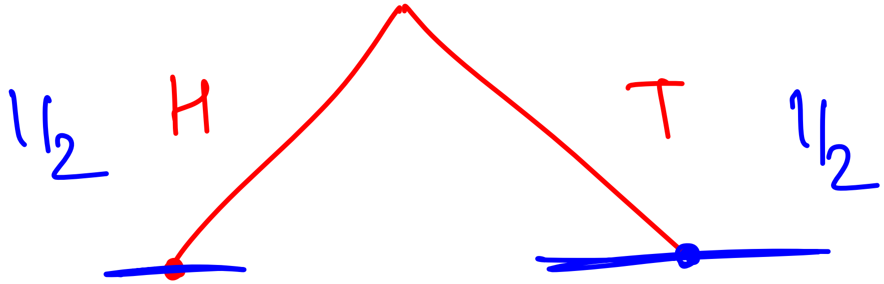
$E(\text{get 1st Head})$



$$E(H) = 1 \cdot 1/2 + 1/2\,(1 + E(H))$$

$$E(H) = 2$$

E(no. of throws to get 1st head)



$1 + \underline{0}$

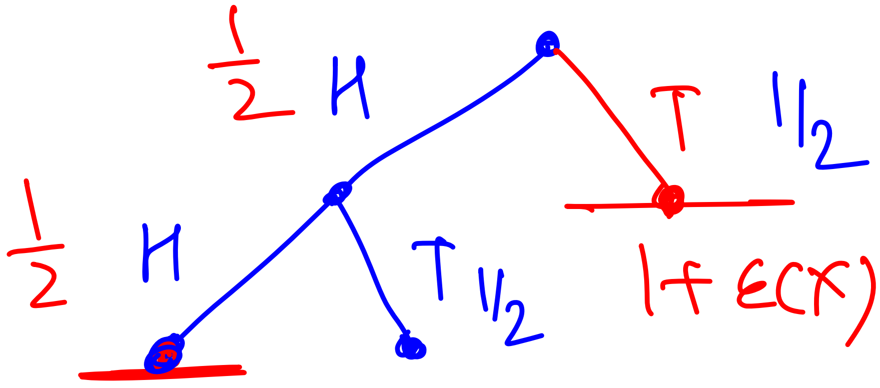
$1 + E$(no. of throws to get 1st head)

$$E(H) = \frac{1}{2}\cdot(1) + \frac{1}{2}\cdot(1 + E(H))$$

Problem 2:

What is expected number of throws after which you will get 2 consecutive heads.

$E(x)$



2

$2 + E(x)$

$$E(x) = \frac{3}{2} + \frac{3}{4}(E(x))$$

$$\frac{1}{4}E(x) = 3/2$$

$$E(x) = 6$$

$$E(x) = \frac{1}{4}(2) + \frac{1}{4}(2 + E(x)) + \frac{1}{2}\cdot(1 + E(x))$$

$$= 1/2 + 1/2 + E(x)/4 + 1/2 + E(x)/2$$

Problem 3:

What is expected number of interviews we will have to take to hire 10 candidates if the probability of getting hired for a candidate is 1/3

$$E(10 \text{ hires}) = E(H_1) + \underline{E(H_2)} + E(H_3)$$

$$\ldots \; E(H_{10})$$

$$E(H_1) = \frac{1}{P(\text{candidate interviewing gets hired})}$$

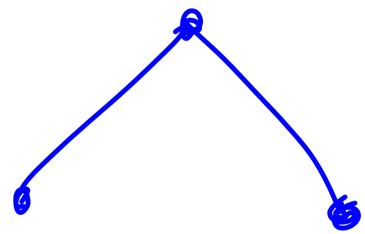
$$= \frac{1}{1/3} = 3$$

$$= \underline{30}$$

$p(\text{A happening}) = p$ $\qquad \underline{E(\text{1st A})}$

$p(\text{A not happening}) = 1-p$

(p) A $\qquad$ A' $\quad (1-p)$



$1 \qquad\qquad 1 + E(\text{1st A})$

$$E(\text{1st A}) = p\cdot 1 + (1-p)\cdot(1+E(\text{1st A}))$$

$$\cancel{E(\text{1st A})} = \cancel{p} + 1 + \cancel{E(\text{1st A})} - \cancel{p} - p\,E(\text{1st A})$$

$$E(\text{1st A}) = 1/p$$

$E(S)$

$1/6 \quad S \qquad\qquad S' \quad (5/6)$

$1 \qquad\qquad 1 + E(S)$

Problem 4: Candy Lottery [CSES]

Link: https://cses.fi/problemset/task/1727

More cool problems on probability and expectation:

https://www.codechef.com/wiki/tutorial-expectation#:~:text=Thus%2C%20the%20expected%20number%20of,two%20consecutive%20heads%20is%206.

E(maximum candies in a distribution)

$$= \underbrace{\sum_{i=1}^{k} i \times P(\text{maximum candies in the distribution} = i)}$$

P(max candies in a distribution = i)

$n = 2$ $\qquad$ $k = 3$

$[1,1] \to 1 \qquad [2,1] \to 2 \qquad [3,1] \to 3$

$[1,2] \to 2 \qquad [2,2] \to 2 \qquad [3,2] \to 3$

$[1,3] \to 3 \qquad [2,3] \to 3 \qquad [3,3] \to 3$

$$1 \cdot \frac{1}{9} + 2 \cdot \frac{3}{9} + 3 \cdot \frac{5}{9}$$

$$P(\text{max candies} = i) = \frac{\{\text{outcomes in which maximum} = i\}}{\text{total outcomes}}$$

k^n

n children, k candies

atleast 1 child got i

and rest got $\leq i$

outcomes in which max = i

1 child got i and rest got < i

+

2 children got i and rest got < i

+

3 child " i " " " < i

+

⋮

n child " i " " " < i

outcomes in which max $= i$

$$= \sum_{j=1}^{n} \text{value of } j \text{ children} = i \text{ and value of } n-j \text{ children} = < i$$

$$= \sum_{j=1}^{n} \underline{\underline{C(n,j)}} \times \underline{\underline{(i-1)^{n-j}}}$$

$$P(\max = i)$$

$$= \frac{\sum_{j=1}^{n} {}^{n}C_{j} \cdot (i-1)^{n-j}}{k^{n}}$$

$$E(\max) = \sum_{i=1}^{k} i \times \left[\frac{\sum_{j=1}^{n} {}^{n}C_{j} \cdot (i-1)^{n-j}}{k^{n}} \right]$$

Outcomes (max = i)

$$\text{Outcomes}(\max \le i) - \text{Outcomes}(\max \le i-1)$$

$$\underbrace{}_{i^n} \qquad\qquad (i-1)^n$$

$$P(\max = i) = \frac{i^n - (i-1)^n}{k^n}$$

$$E(\max) = \sum_{i=1}^{k} i \times \left(\frac{i^n - (i-1)^n}{k^n} \right)$$

Problems based on Bounds

Majority Element Problem

(10, 9, 10, 10, 8, 10)

n = 6

10 → 4

> 6/2

Given an array, find the majority element in it. Given that the majority element always exists. Majority Element = element occuring more than n/2 times in the array.

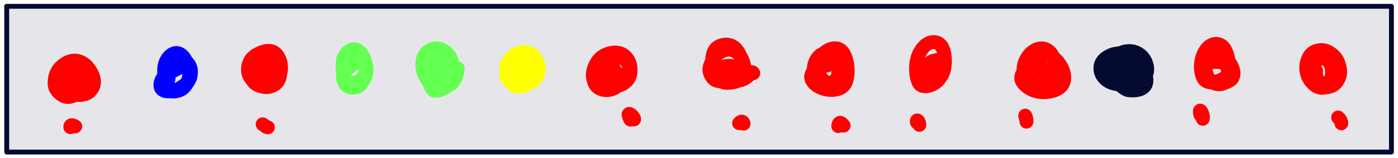
in O(1) space

$1 <= n <= 10^5$

$1 <= arr[i] < 10^{18}$

O(n)

> n/3



$$P(R) = 9/14$$

majority element $\longrightarrow$ $> n/2$

$$P(\text{majority element}) \longrightarrow \frac{\text{occ}(\text{majority element})}{n}$$

$$\longrightarrow \frac{> n/2}{n} \longrightarrow \underline{\underline{> 1/2}}$$

$$P(M) = > 1/2$$

$$P(M') = 1 - P(M)$$
$$= 1 - (> 1/2)$$
$$= < 1/2$$

$$P(M_1') = < 1/2$$

$$P(M_1' \cdot M_2') = P(M_1') \cdot P(M_2')$$
$$= < 1/2 \cdot < 1/2 = < 1/4$$

$$P(M_1' \cdot M_2' \cdot M_3') = < 1/8$$

$$P(M_1' \cdot M_2' \cdot M_3' \ldots \ldots M_k') = < 1/2^k$$

$k = 10$

$$< 1/1024 = < 0.001$$

$k = 20$ $\quad < 1/10^6 = < 0.000001$

$k = 30$ $\quad < 1/10^9 < 0.000000001$

$k = 10 \rightarrow < 0.001 \rightarrow$ WA

$k = 10$ 1st submission < 0.001 WA

$k = 10$ 2nd sub

$P($ 1st system fails$)$ $= < 0.001$

$P($ 1st failed and 2nd also fails$)$

$= < 0.001 \times 0.001$

$= < 0.000001$

Problems based on Bounds

Good Line Segment Problem

Given a list of N points, find a line segment that passes through at least N/3 points. Given that the answer always exists.

$$3 <= n <= 10^3$$
$$1 <= x[i], y[i] < 10^{18}$$

Bonus:

$$3 <= n <= 10^5$$
$$1 <= arr[i] < 1e^{18}$$